

weighs less than 150 milligrams. It is smaller and lighter than other active-cooling solutions. The chip can move up to 39 cubic centimetres of air per second, achieving 1000Pa of back pressure. This all-silicon solution provides semiconductor reliability, uniformity across parts, durability, and is IP58 rated.

xMEMS Labs
<https://xmems.com>

4-in-1 MEMS air quality sensor for indoors

Bosch Sensortec has introduced the BME690, a gas sensor that tracks gases, temperature, pressure, and humidity. This sensor builds on the technology of its predecessors, the BME688 and



BME680, maintaining the same dimensions for easy integration as an upgrade. The sensor monitors indoor air,

detecting volatile organic compounds (VOCs) from cleaners, cooking, and other sources. It is also used for analysing breath and detecting food spoilage, helping reduce waste. Equipped with AI features through Bosch Sensortec's BME AI-Studio software, the sensor allows customers to create solutions swiftly, giving them a competitive edge. The sensor also reduces power consumption by up to 50% compared to its predecessor, making it suitable for portable devices such as wearables and asset trackers.

Bosch Sensortec
<https://www.bosch-sensortec.com>

World's first single chip solution for speech recognition

Applied Brain Research (ABR) has launched the world's first single-chip speech recognition system. This introduces the TSP1, ABR's first time series processor chip, designed for real-time, low latency automatic speech recognition. It starts with the Legendre memory unit (LMU; patented), the first state-space network, which optimises

computation for time series processing. The networks are then trained and compiled using a toolchain, and run



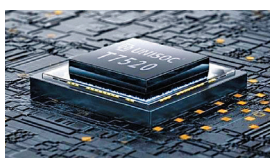
on a neural fabric, which reduces power consumption by minimising data move-

ment within the chip. The TSP1 is a single-chip solution for time series inference applications such as real-time speech recognition, text-to-speech synthesis, natural language interfaces, and sensor fusion.

Applied Brain Research
<https://www.appliedbrainresearch.com>

World's first open architecture RISC-V security chip

Unisoc has launched what it claims to be the world's first security chip based on the RISC-V open instruction set architecture (ISA). The E450R chip is reported to enhance the performance



of security algorithms, according to IT Home. A feature of the chip

is its asymmetric cryptographic PKE algorithm engine, which reportedly operates 50% faster than its predecessor. This improvement is maintained even as key lengths are increased, ensuring security without compromising on speed. The chip also shows a 50% increase in performance for typical transaction-based applications, making it suitable for environments that require quick processing. Additionally, the chip has a 15% increase in the erase/write speed of its non-volatile memory (NVM), enabling more efficient data storage in applications that frequently perform memory operations. Moreover, the chip's microarchitecture not only reduces the size of application code by 30% but also doubles the speed of application loading, which is important for real-time applications.

Unisoc
https://www.unisoc.com/en_us

Chip with dual core and dual architecture

Raspberry Pi has introduced a micro-controller, the RP2350. The chip integrates two ARM Cortex M33 cores with a floating point unit and two



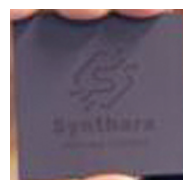
RISC-V HAZARD3 cores, though only two of these cores can be active simultaneously. Its clock speed has been increased to 150MHz. Thanks

to its new architecture, floating point unit, and three PIO units, the chip achieves roughly double the performance of its predecessors at standard clock speeds while consuming less power. The RP2350 features a dual-core, dual-architecture design, offering two industry-standard ARM Cortex-M33 cores and two open-hardware Hazard3 RISC-V cores. These can be selected either through software settings or by programming the on-chip OTP memory.

Raspberry Pi
<https://www.raspberrypi.org>

Network-on-chip accelerator

Synthara has completed the first tape-out of the network-on-chip accelerator, Watchdog (WDG). The team developed



WDG in response to the demands of AI and machine learning. It connects multiple processing elements, enabling them to communi-

cate and share data effectively. This integration leads to quicker and more efficient data transfers, improving the performance of AI models. A key feature of Watchdog is its compute-in-memory capability, which minimises data movement and latency by processing data directly where it is stored. This approach allows for faster processing speeds, significantly enhancing computation capabilities.

Synthara
<https://synthara.ai>